Assignment 1 - Git & GitHub Fundamentals

Session 1 | Group D | Individual Lab Work

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Repository: https://github.com/KBLLL05/git-session1-KBLLL05

1. Repository Setup

The supplied starter HTML and CSS were imported into git-session1-KBLLL05, then git init -b main initialized the repository. The .gitignore excludes local.env, .env, and *.log; !important.log keeps the required log visible. git check-ignore -v verified the rules, and git ls-files confirmed that important.log is tracked. A .gitattributes rule treats PDF files as binary to preserve their bytes across platforms.

2. README & Commit History

The README contains my name, GitHub username, project description, HTML/CSS/Git/Markdown technologies, a shell code block, and links. The AUI logo was extracted from the README example on lecture slide 12 because the starter ZIP did not contain a separate image. The initial work was split into five focused commits: starter setup, README, homepage structure, styling, and the About feature. Later commits record the merge, report preparation, identity/logo updates, and remote synchronization.

3. Selective Staging

index.html and style.css were modified together. git diff showed unstaged edits. git add index.html staged only the homepage; git diff --cached showed only that file. Commit 2dea208 recorded the homepage while git status --short still showed style.css modified. The stylesheet was committed separately as 13a8d16. This demonstrates that a commit records the index, not every change in the working directory.

4. Branch & Merge

git switch -c feature/about created the feature branch. Commit 38d8167 added about.html and a homepage link. After git switch main, git merge --no-ff feature/about created merge commit a637cda. The graph on page 3 shows that the feature branch was merged into main.

5. Remote Changes

An empty public GitHub repository was created, connected using git remote add origin, and published with git push -u origin main. A commit saves a staged snapshot locally; push uploads commits to a remote. The README was edited directly in the GitHub browser editor, creating f36e02a. git fetch origin updated origin/main while leaving local HEAD and the README unchanged. git status -sb showed [behind 1], and git rev-list --left-right --count HEAD...origin/main returned 0 1. git pull --ff-only origin main then integrated that commit; the comparison became 0 0. Fetch updates remote-tracking information; pull fetches and integrates it into the current branch.

6. Fair Use of AI

I used AI to check whether there were any improvements I could make to my report.

7. Sources & Reflection

Sources: Assignment 1 and Report Template - Group D.pdf (requirements and report headings); git presentation - session 1 - group D.pdf (Git workflow and slide 12 logo); Session1_Practice - group D.zip (starter HTML, CSS, and practice logs). The most subtle concept in this workflow is separating the working tree, staging area, and repository history: a modified file stays outside a commit until it is staged. When Git behaves unexpectedly, the first checks are git status, the current branch, and git diff versus git diff --cached.

Troubleshooting

A controlled git add app.log attempt failed because *.log is ignored. git check-ignore -v app.log identified the exact rule; the file was correctly left untracked instead of forcing it into Git. Network commands also initially failed in the restricted execution environment; after network approval, push, fetch, and pull succeeded without changing repository URLs or ignore rules.

Evidence - recorded Git states

These are excerpts of real command output. Full transcripts are in evidence/local-git-evidence.txt and evidence/remote-git-evidence.txt.

```
$ git status --short  # after staging only index.html
M  index.html
 M style.css

$ git status --short  # after commit 2dea208
 M style.css

$ git check-ignore -v app.log debug.log local.env
.gitignore:3:*.log       app.log
.gitignore:3:*.log       debug.log
.gitignore:1:local.env   local.env

$ git status --short --ignored  # after local feature merge
!! app.log
!! debug.log
!! local.env
```

After fetch, before pull: local HEAD stayed at a58110e. The README blob hash before and after fetch was fb039e60cc268b369921e877c90732a1b7d22874.

```
$ git status -sb
## main...origin/main [behind 1]
$ git rev-list --left-right --count HEAD...origin/main
0  1
$ git log --oneline HEAD..origin/main
f36e02a Document remote synchronization in GitHub editor

$ git pull --ff-only origin main
Updating a58110e..f36e02a
Fast-forward
 README.md | 2 ++
 1 file changed, 2 insertions(+)
$ git rev-list --left-right --count HEAD...origin/main
0  0
```

History through the GitHub README edit (the final report commit follows this snapshot):

```
* f36e02a Document remote synchronization in GitHub editor
* a58110e Add student details and university logo from lecture
* ae18643 Preserve PDF report as binary across platforms
* 79d53e8 Add draft lab report and verified local workflow evidence
*   a637cda Merge About feature into main
|\
| * 38d8167 Add About page and homepage navigation
|/
* 13a8d16 Improve page spacing colors and readability
* 2dea208 Structure homepage and connect stylesheet
* 673d31b Document project purpose technologies and learning goals
* ab536b2 Initialize starter project and ignore local configuration
```

Evidence - README rendered on GitHub

The browser screenshot shows the GitHub commit, rendered Markdown, student name, technologies, code block, university logo, and remote synchronization sentence.

